Prescribed digit sums in residue classes
and a conjecture of Murthy (OEIS A077754)

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Abstract

Let $m \geq 2$ be coprime to 10, and put $g = \gcd(9, m)$ and $M = m/g$. Fix a decimal length $L \geq 1$ and set $s = \lceil (M-1)/2 \rceil$. We prove that, whenever $L \geq 3s$, *every* digit sum σ in the band $[L + 3s,\ 9L - 3s]$ and *every* residue $r \equiv \sigma \pmod{g}$ is realized by an L-digit positive integer N with all digits nonzero, $\sigma(N) = \sigma$, and $N \equiv r \pmod{m}$; this band is the feasibility window of the balanced exchange method. The same machinery, run with denser smooth moves and finite certificates in place of minimal moves, resolves the conjecture of Amarnath Murthy recorded as OEIS A077754 (2002): a positive integer $n \geq 1$ admits a multiple with at least two digits, all nonzero, and digit sum n, if and only if $n \notin \{1, 2, 5, 11\} \cup \{10k : k \geq 1\}$. Both implications are machine-verified in full in Lean 4 with mathlib: all finite lemmas are kernel-checked by decide on explicit witness tables and bitmap certificates, so the axiom base reduces to the three standard axioms (propext, Classical.choice, Quot.sound), and an independent self-contained Python program re-derives the main finite checks in seconds. The OEIS sequence value is formalized in a companion module, certifying that it is the least admissible multiple for every non-exceptional $n \geq 1$, vanishes on the exceptional set, and agrees with the displayed OEIS values. The proof is elementary modulo one classical sumset theorem of Chowla (1935).

1 Statement of results

Throughout this paper, $\mathbb{N} = \{0, 1, 2, \dots\}$ denotes the natural numbers, and $10\mathbb{N} = \{10k : k \in \mathbb{N}\}$ denotes the set of multiples of 10. For a positive integer $N \geq 1$, we write $\sigma(N)$ for the sum of the decimal digits of N (with the convention $\sigma(0) = 0$) and $\mathrm{len}(N)$ for its number of decimal digits.

Entry A077754 in the On-Line Encyclopedia of Integer Sequences [1], contributed by Amarnath Murthy in November 2002 for indices $n \geq 1$, reads verbatim:

> "Smallest multiple of n with two or more digits, none of them zeros, whose digit sum equals n, or 0 if no such multiple exists. Conjecture: zero occurs only for indices which are multiples of 10 apart from 1, 2, 5 and 11. (a(n) = 0 only for n = 1,2,5,11 or n = 10k.)"

Our first result establishes residue coverage across a wide band of digit sums on fixed-length decimal fibres without zero digits. Murthy's conjecture is resolved by the same balanced-exchange machinery, using denser smooth moves and finite certificates where the minimal-move band does not reach $\sigma = n$.

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Theorem 1 (equidistribution on the digit fibre)**.** *Let $m \geq 2$ be coprime to 10, and set $g = \gcd(9, m)$, $M = m/g$, and $s = \lceil (M-1)/2 \rceil$. For every positive integer $L \geq 3s$, every integer $\sigma \in [L + 3s,\ 9L - 3s]$, and every residue $r \equiv \sigma \pmod{g}$, there exists an L-digit positive integer N with all decimal digits nonzero such that $\sigma(N) = \sigma$ and $N \equiv r \pmod{m}$. The congruence $r \equiv \sigma \pmod{g}$ is necessary because $N \equiv \sigma(N) \pmod{g}$, and $[L + 3s,\ 9L - 3s]$ is the feasibility window certified by the balanced exchange method (§7).*

Theorem 2 (Murthy's conjecture, OEIS A077754 [1])**.** *A positive integer $n \geq 1$ admits a multiple N with at least two decimal digits, all nonzero, and $\sigma(N) = n$, if and only if $n \notin \{1, 2, 5, 11\} \cup \{10k : k \geq 1\}$.*

Theorem 2 is formally verified in Lean 4 with mathlib (v4.35.0) [7] without `sorry` or `native_decide`, relying only on the three standard axioms `propext`, `Classical.choice`, and `Quot.sound` (see §8 for the exact distinction between kernel-formalized theorems and paper-only band arguments). Note that on $\mathbb{N}$, the Lean statement includes $n = 0 \in 10\mathbb{N}$, which is vacuously excluded because $\text{len}(0) < 2$; restricting to $n \geq 1$ gives Theorem 2.

The necessity of the exceptions is elementary (§2). For sufficiency, the range $1 \leq n \leq 200$ is settled by an exhaustive witness table (certificate A in the appendix). For $n > 200$, we write $n = R \cdot m$ where R is the $\{2, 5\}$-part of n and $\gcd(m, 10) = 1$. We concatenate a short low-order block B divisible by R with a high-order prefix H drawn from a fixed-length digit fibre, realizing the target residue $-B \cdot 10^{-d} \pmod{m}$ (where negative exponents denote modular inverses modulo m) via balanced digit exchanges. We prove Theorem 2 first using dense move sets tailored to $n > 200$ (§5), and then apply the minimal move set to establish the general band theorem (§6).

Corollary 3 (quantitative length bound)**.** *For every non-exceptional $n > 200$, the constructed witness N satisfies $\text{len}(N) \leq \lfloor n/5 \rfloor + \lfloor \log_2 n \rfloor$: it consists of a high fibre of $L = \lfloor (n - \sigma(B))/5 \rfloor$ digits (for $m \geq 2$; in the $m = 1$ branch the fibre prefix has $\lceil (n - \sigma(B))/9 \rceil \leq L$ digits) concatenated with a low block of $d \leq \log_2 n$ (or $d \leq \log_5 n$) digits.*

The bound in Corollary 3 is existential; the *smallest* admissible multiple – the OEIS sequence value $a(n)$ itself – is formalized and certified separately for the displayed anchor values in §8.

Related work. When zero digits are permitted, the existence of a multiple of n with digit sum n is classical (see OEIS A002998 [2] and Aassila [6]). Related results on digit sums of multiples and minimal Niven numbers include Fredricksen, Ionascu, Luca, and Stănică [5]. None of these works enforces the all-nonzero-digit constraint, which creates the exact exceptional set $\{1, 2, 5, 11\} \cup \{10k : k \geq 1\}$ in Theorem 2.

2 Necessity of the exceptions

We verify that no admissible multiple N exists when $n \geq 1$ lies in $\{1, 2, 5, 11\} \cup \{10k : k \geq 1\}$.

- If $10 \mid n$, then every multiple N of n ends in the decimal digit 0, violating the nonzero-digit requirement.
- If $n = 1$, every integer with at least two nonzero digits satisfies $\sigma(N) \geq 2 > 1$.
- If $n = 2$, any multiple N of 2 with all digits nonzero must end in an even nonzero digit (at least 2). Having at least two nonzero digits then forces $\sigma(N) \geq 1 + 2 = 3 > 2$.
- If $n = 5$, any multiple N of 5 with all digits nonzero must end in 5. Having at least two nonzero digits forces $\sigma(N) \geq 1 + 5 = 6 > 5$.

- If $n = 11$, write $S = \sigma(N)$ for the digit sum of an admissible multiple N and A for the alternating sum of its digits. Suppose $S = 11$ and $11 \mid N$. Then $11 \mid A$, and $A \equiv S = 11 \pmod 2$, so A is odd. Moreover, $|A| \le S = 11$. If $|A| = 11$, then all digits in either the even or the odd decimal positions must vanish; since all digits are nonzero, N can have at most one digit, contradicting $\mathrm{len}(N) \ge 2$. Thus $|A| < 11$, which is impossible because no odd integer strictly between -11 and 11 is divisible by 11. □

These cases reflect three distinct obstructions: terminal digit divisibility ($10k$ and 5), minimum digit-sum bounds (1 and 2), and alternating digit weight (11).

3 Machinery: the fixed-count digit fibre

Let $n \ge 1$ be a positive integer not divisible by 10. Write $n = R \cdot m$ with $\gcd(m, 10) = 1$ and $R \in \{1, 2^a, 5^b\}$ for integers $a, b \ge 1$; the primes 2 and 5 cannot both divide n because $10 \nmid n$. When $m \ge 2$, set $g = \gcd(9, m) \in \{1, 3, 9\}$ and $M = m/g$.

Our witness has the form $N = B + 10^d \cdot H$, where B is a d-digit low-order block divisible by R and H is an L-digit high-order prefix whose residue modulo m we control by balanced exchanges. Because $B < 10^d$ has exactly d nonzero digits (when $d \ge 1$), the decimal representation of N is the concatenation of H and B, so no carries cross the interface and $\sigma(N) = \sigma(H) + \sigma(B)$.

The digit fibre. Fix a length $L \ge 1$ and an offset sum $K \in \{0, \dots, 8L\}$. Every offset vector $u = (u_0, \dots, u_{L-1})$ with $u_e \in \{0, \dots, 8\}$ and $\sum_{e=0}^{L-1} u_e = K$ encodes an L-digit integer

$$F(u) = \sum_{e=0}^{L-1} (1 + u_e)\, 10^e.$$

Because each digit $1 + u_e$ lies in $\{1, \dots, 9\}$, no carries occur in base 10, all L digits of $H = F(u)$ are nonzero, $\sigma(H) = L + K$, and $H \equiv L + K \pmod g$ since $10^e \equiv 1 \pmod 9$ and $g \mid 9$.

Lemma 4 (the low block)**.** *If $R = 1$, set $d = 0$ and $B = 0$. If $R = 2^a$ with $a \ge 1$, there exists an a-digit integer B with all digits in $\{1, 2\}$ such that $2^a \mid B$ and $\sigma(B) \le 2a$. If $R = 5^b$ with $b \ge 1$, there exists a b-digit integer B with all digits in $\{1, 2, 3, 4, 5\}$ such that $5^b \mid B$ and $\sigma(B) \le 5b$.*

Proof. For $R = 2^a$, proceed by induction on a. For $a = 1$, take $B = 2$. Given a j-digit multiple B_j of 2^j with digits in $\{1, 2\}$, we have $10^j = 2^j \cdot 5^j \equiv 2^j \pmod{2^{j+1}}$, so one of $B_j + 1 \cdot 10^j$ or $B_j + 2 \cdot 10^j$ is divisible by 2^{j+1}. Similarly, for $R = 5^b$, take $B_1 = 5$; given $5^j \mid B_j$, write $10^j = 5^j \cdot 2^j$. Since $\gcd(2^j, 5) = 1$, as c ranges over $\{1, 2, 3, 4, 5\}$ the values $B_j + c \cdot 10^j$ cover all five multiples of 5^j modulo 5^{j+1}, so one is divisible by 5^{j+1}. The digit-sum bounds follow immediately. □

Balanced exchanges. On a block of three consecutive positions $\{3i, 3i + 1, 3i + 2\}$ inside $\{0, \dots, L - 1\}$, consider a triple $(d_0, d_1, d_2) \in \{-4, \dots, 4\}^3$ such that $d_0 = -d_1 - d_2$ and $u_{3i+r} + d_r \in \{0, \dots, 8\}$ for $r \in \{0, 1, 2\}$. Adding (d_0, d_1, d_2) to $(u_{3i}, u_{3i+1}, u_{3i+2})$ preserves the total sum $\sum_e u_e = K$, keeps all digits in $\{1, \dots, 9\}$, and shifts the value $F(u)$ by

$$\Delta = (d_0 + 10d_1 + 100d_2) \cdot 10^{3i} = 9 \cdot 10^{3i} \cdot (d_1 + 11d_2).$$

When $u_{3i} = u_{3i+1} = u_{3i+2} = 4$, any triple with $d_1, d_2, -d_1 - d_2 \in \{-4, \dots, 4\}$ is admissible; the resulting integer grid

$$X = \{\, d_1 + 11d_2 : d_1, d_2 \in \{-4, \dots, 4\},\ |d_1 + d_2| \le 4 \,\}$$

contains 61 elements (0 together with 60 nonzero integers, all bounded in absolute value by 44). When we only assume $u_{3i+r} \in \{1, \ldots, 7\}$ for $r \in \{0, 1, 2\}$, restricting to $d_0, d_1, d_2 \in \{-1, 0, 1\}$ with $d_0 = -d_1 - d_2$ yields $d_1 + 11d_2 \in \{0, \pm 1, \pm 10, \pm 11\}$; in particular, the *minimal moves* $x \in \{\pm 1, \pm 10\}$ are always available on any triple of positions with offsets in $\{1, \ldots, 7\}$.

Frozen poles. An offset $u_e = 0$ (digit 1) admits only $d_r \geq 0$, and an offset $u_e = 8$ (digit 9) admits only $d_r \leq 0$. If all three positions of a block sit at $u = 0$ (or all three sit at $u = 8$), the balance condition $d_0 + d_1 + d_2 = 0$ forces $d_0 = d_1 = d_2 = 0$. Requiring every position in each active exchange block to satisfy $u_e \in \{1, \ldots, 7\}$ avoids both frozen poles uniformly.

4 Stage sets and the Chowla step

Let $m \geq 2$ be coprime to 10, $g = \gcd(9, m)$, and $M = m/g$. Because $\gcd(m, 10) = 1$, M is coprime to 10 (in particular, M is odd). Moreover, $\gcd(9/g, M) = 1$: this is trivial for $g \in \{1, 9\}$, and if $g = 3$ then $9 \nmid m$, so $3 \nmid (m/3) = M$.

If $M = 1$ (which occurs when $m \in \{3, 9\}$), the ring $\mathbb{Z}/M\mathbb{Z}$ is trivial: every integer is congruent to 0 (mod M), so $s = 0$ stages already cover $\mathbb{Z}/M\mathbb{Z}$ once the congruence modulo $g = m$ holds.

Assume henceforth that $M \geq 2$ (and hence $M \geq 3$, since $\gcd(M, 10) = 1$). For a subset $U \subseteq X$ of grid values that are admissible on block i, we define the *stage set* modulo M by

$$B_i^*(U) = \{0\} \cup \{\, (9/g) \cdot 10^{3i} \cdot x \bmod M : x \in U,\ \gcd(x, M) = 1 \,\} \subseteq \mathbb{Z}/M\mathbb{Z}.$$

Because $\gcd(9/g, M) = \gcd(10^{3i}, M) = 1$, multiplication by $(9/g) \cdot 10^{3i}$ is a unit automorphism of $\mathbb{Z}/M\mathbb{Z}$: every nonzero element of $B_i^*(U)$ is a unit in $\mathbb{Z}/M\mathbb{Z}$, and $|B_i^*(U)| = |B_0^*(U)|$ is independent of i. Distinct blocks i may produce the same subset of $\mathbb{Z}/M\mathbb{Z}$ (for instance when $10^3 \equiv 1$ (mod M), as for $M = 333$), which causes no difficulty because different blocks act on disjoint triples of decimal positions.

Theorem 5 (Chowla, 1935 [3])**.** *Let $k \geq 2$ be an integer, and let $A, B \subseteq \mathbb{Z}/k\mathbb{Z}$ be nonempty subsets. If $0 \in B$ and every nonzero element of B is coprime to k, then $|A + B| \geq \min(k,\ |A| + |B| - 1)$.*

Applying Theorem 5 iteratively across s distinct position blocks $i_1, \ldots, i_s$ (which requires $3s$ positions in total), starting from $\{0\} \subseteq \mathbb{Z}/M\mathbb{Z}$, yields

$$|B_{i_1}^*(U) + \cdots + B_{i_s}^*(U)| \geq \min(M,\ s \cdot w + 1), \qquad w = |B_0^*(U)| - 1.$$

Hence $s = \lceil (M - 1)/w \rceil$ stages cover all of $\mathbb{Z}/M\mathbb{Z}$ whenever $w \geq 1$. We apply this in two regimes (still assuming $M \geq 3$):

- *Full smooth moves* (base offsets $u_e = 4$ on the active blocks, admitting all of X): when $g \in \{3, 9\}$, the $\{2, 5\}$-smooth elements of $X \setminus \{0\}$ are

  $$U_{2,5} = \{\pm 1, \pm 2, \pm 4, \pm 8, \pm 10, \pm 20, \pm 32, \pm 40\},$$

  a set of 16 integers automatically coprime to M since $\gcd(M, 10) = 1$. When $g = 1$, we also have $3 \nmid M$, so we may adjoin the $\{2, 3, 5\}$-smooth elements $\pm\{3, 9, 12, 18, 24, 30\} \subseteq X$, obtaining a 28-element set $U_{2,3,5} \subseteq X$ of integers coprime to M (certificate C). All elements of $U_{2,5}$ and $U_{2,3,5}$ lie in $[-40, 40]$, so their pairwise differences are nonzero and bounded in absolute value by 80. Therefore, *whenever $M > 80$*, reduction modulo M is injective on $U_{2,5}$ and on $U_{2,3,5}$ (and none of these units can reduce to 0 mod M), giving $w = |B_0^*(U_{2,5})| - 1 = 16$

for $g \in \{3, 9\}$ and $w = |B_0^*(U_{2,3,5})| - 1 = 28$ for $g = 1$. Consequently, for $M > 80$, at most $s = \lceil (M-1)/w \rceil$ stages cover $\mathbb{Z}/M\mathbb{Z}$. (The Lean formalization uses the slightly more conservative uniform step count $s = \lfloor m/27 \rfloor + 1$ – that is, $\lfloor M/27 \rfloor + 1$, $\lfloor M/9 \rfloor + 1$, or $\lfloor M/3 \rfloor + 1$ according as $g = 1$, 3, or 9 – which satisfies the same $3s + 4 \le L$ inequality for all $M > 184$.)

- *Minimal moves* (offsets $u_e \in \{1, \ldots, 7\}$ on the active blocks): take $U_{\min} = \{\pm 1, \pm 10\}$. Since $\gcd(M, 10) = 1$, all four elements of $U_{\min}$ are units modulo M. Because $M \ge 3$ is odd and $c = (9/g) \cdot 10^{3i}$ is a unit modulo M, the residues $+c$ and $-c$ are distinct modulo M (otherwise $M \mid 2c$, impossible since $\gcd(2c, M) = 1$). Thus $B_i^*(U_{\min}) \supseteq \{0, +c, -c\}$ satisfies $|B_i^*(U_{\min})| \ge 3$, so $w \ge 2$ for every $M \ge 3$, and $s = \lceil (M-1)/2 \rceil$ stages cover $\mathbb{Z}/M\mathbb{Z}$. (In fact, adding $\{0, +c, -c\}$ to any nonempty proper subset $A \subsetneq \mathbb{Z}/M\mathbb{Z}$ strictly increases A by at least 2 elements up to size M by considering orbits of $+c$, giving Theorem 5 directly in this special case.)

5 Proof of Theorem 2

Necessity was proved in §2, and every non-exceptional n in the range $1 \le n \le 200$ is resolved by an explicit witness in certificate A. Assume now that $n > 200$ and $10 \nmid n$. Write $n = R \cdot m$ with $\gcd(m, 10) = 1$ and $R \in \{1, 2^a, 5^b\}$, and let B and d be given by Lemma 4.

Case $m = 1$. Here $n = R > 200$. If $n = 2^a$, then $a \ge 8$ and $\sigma(B) \le 2a < 2^a = n$. If $n = 5^b$, then $b \ge 4$ and $\sigma(B) \le 5b < 5^b = n$. Thus $S := n - \sigma(B) \ge 1$ in both cases. Write $S = 9q + r$ with $q \ge 0$ and $r \in \{1, \ldots, 9\}$, and let H be the integer consisting of the digit r followed by q digits equal to 9. All digits of H are nonzero, $\sigma(H) = S$, and $N = B + 10^d \cdot H$ has all digits nonzero, $\mathrm{len}(N) \ge d + 1 \ge 5 \ge 2$, $\sigma(N) = \sigma(B) + S = n$, and $R \mid N$ since $R \mid B$ and $R \mid 10^d$.

Case $m \ge 2$: target and coset compatibility. Since $\gcd(m, 10) = 1$, 10^d is invertible modulo m; set the target residue $t := -B \cdot 10^{-d} \bmod m$ (so $t = 0$ when $R = 1$). We choose the fibre parameters

$$L = \Big\lfloor \frac{n - \sigma(B)}{5} \Big\rfloor, \qquad K = n - \sigma(B) - L = 4L + r_0, \quad r_0 \in \{0, 1, 2, 3, 4\},$$

provided $n - \sigma(B) \ge 5$ (verified below). Because $B \equiv \sigma(B) \pmod 9$ and $10^d \equiv 1 \pmod 9$, we have $t \equiv -\sigma(B) \pmod g$. On the fibre, every encoded value $F(u)$ satisfies $F(u) \equiv L + K = n - \sigma(B) \equiv -\sigma(B) \equiv t \pmod g$, because $g \mid m$ and $m \mid n$. Thus $(t - F(u^0))/g$ is an integer for any base offset vector u^0 of sum K.

To place s disjoint exchange blocks at base offset $u_e = 4$ and distribute the remainder $r_0 \le 4$ by raising r_0 separate positions from 4 to 5, it suffices that $L \ge 3s + 4$ (or $L \ge 3s + r_0$).

Bounding $\sigma(B)$ and $n - \sigma(B)$. If $R = 1$, then $\sigma(B) = 0$ and $n - \sigma(B) = m = gM$. Suppose $R \ge 2$. If $R = 2^a$ ($a \ge 1$), then $2^a - 2a \ge 0$ for all $a \ge 1$; if $R = 5^b$ ($b \ge 1$), then $5^b - 5b \ge 0$ for all $b \ge 1$. Moreover, since $m \ge 3$ (as $\gcd(m, 10) = 1$ and $m \ge 2$), we have

$$n - \sigma(B) = Rm - \sigma(B) = m + (R-1)m - \sigma(B) \ge gM + 3(R-1) - \sigma(B).$$

For $R = 2^a$ ($a \ge 1$), $3(2^a - 1) - 2a \ge 3 \cdot 1 - 2 = 1 > 0$; for $R = 5^b$ ($b \ge 1$), $3(5^b - 1) - 5b \ge 3 \cdot 4 - 5 = 7 > 0$. Therefore, in *all* cases ($R = 1$ or $R \ge 2$), we have the clean lower bound

$$n - \sigma(B) \ge m = gM.$$

Similarly, because $n > 200$, one checks directly that $n - \sigma(B) \geq 188$: for $R = 1$ we have $n - \sigma(B) = n \geq 201$; for $R = 2^a$ with $n = 2^a m > 200$ and $m \geq 3$, we have $2^a m - 2a \geq 202 - 14 = 188$ (and if $a \geq 8$, $2^a m - 2a \geq 3 \cdot 256 - 16 = 752$); for $R = 5^b$ with $n = 5^b m > 200$ and $m \geq 3$, the smallest value of $5^b m - 5b$ for $b \geq 2$ occurs at $b = 2, m = 9$ ($n = 225$), giving $225 - 10 = 215$ (the next case $b = 3, m = 3$ gives $375 - 15 = 360$, and $b \geq 4$ gives ≥ 1855), while $b = 1$ has $m \geq 41$ and $5m - 5 \geq 200$. Hence $n - \sigma(B) \geq 188$ for every $n > 200$ with $m \geq 2$, which forces

$$L = \left\lfloor \frac{n - \sigma(B)}{5} \right\rfloor \geq \left\lfloor \frac{188}{5} \right\rfloor = 37.$$

Verifying the stage budget $3s + 4 \leq L$. We split on the size of M.

- *Analytic regime $M > 184$.* Since $M > 184 > 80$, the smooth-unit injectivity holds (§4), giving $w = 28$ when $g = 1$ and $w = 16$ when $g \in \{3, 9\}$, with $s = \lceil (M-1)/w \rceil$. Using $L = \lfloor (n - \sigma(B))/5 \rfloor \geq \lfloor gM/5 \rfloor \geq (gM - 4)/5$:
  - For $g = 1$ ($w = 28$, $M \geq 185$): we need $3\lceil (M-1)/28 \rceil + 4 \leq \lfloor M/5 \rfloor$. Since $\lceil (M-1)/28 \rceil \leq (M + 26)/28$, we have
    $$3\left\lceil \frac{M-1}{28} \right\rceil + 4 \leq \frac{3M + 190}{28} \leq \frac{M - 4}{5}$$
    for all $M \geq 184$ (as $15M + 950 \leq 28M - 112 \iff 13M \geq 1062 \iff M \geq 82$).
  - For $g = 3$ ($w = 16$, $M \geq 185$): we have
    $$3\left\lceil \frac{M-1}{16} \right\rceil + 4 \leq \frac{3M + 106}{16} \leq \frac{3M - 4}{5}$$
    for all $M \geq 185$ (as $15M + 530 \leq 48M - 64 \iff 33M \geq 594 \iff M \geq 18$).
  - For $g = 9$ ($w = 16$, $M \geq 185$): we have $3\lceil (M-1)/16 \rceil + 4 \leq (3M-4)/5 < (9M-4)/5 \leq L$.

  Thus $3s + 4 \leq L$ holds for all $M > 184$.

- *Certified finite regime $M \leq 184$.* Here $m = gM \leq 9 \cdot 184 = 1656$. If $M = 1$ ($m \in \{3, 9\}$), $s = 0$ stages suffice and $4 \leq 37 \leq L$. For $2 \leq M \leq 184$ with $m \notin \{11, 19, 23\}$, certificate B verifies directly by bit-parallel sumset evaluation that the smooth moves $U_{2,3,5}$ (28 values, used when $g = 1$) or $U_{2,5}$ (16 values, used when $g \in \{3, 9\}$) cover $\mathbb{Z}/m\mathbb{Z}$ (within the mod g coset) in $k \leq \lfloor m/27 \rfloor + 1$ stages. Using $n - \sigma(B) \geq m$ and $L \geq 37$: if $m \leq 296$, then $k \leq 11$ and $3k + 4 \leq 37 \leq L$; if $m \geq 297$, then
  $$3k + 4 \leq 3\left(\frac{m}{27} + 1\right) + 4 = \frac{m}{9} + 7 \leq \frac{m - 4}{5} \leq L$$
  where the last step uses the stronger available bound $L \geq \lfloor m/5 \rfloor$ since $5m + 315 \leq 9m - 36 \iff 4m \geq 351 \iff m \geq 88$. Finally, for the three small moduli $m \in \{11, 19, 23\}$, certificate D1 verifies that $k = 2$ stages cover $\mathbb{Z}/m\mathbb{Z}$ (requiring $3k + 4 = 10 \leq 37 \leq L$). (In the Lean formalization, the standalone lemma for $m \geq 2$ is stated more generally for all $L \geq \lfloor m/5 \rfloor$ rather than only $L \geq 37$; for $m \in \{11, 19, 23\}$ at the small lengths $L \in \{7, 8, 9\}$, where $3k + 4 = 10 > L$, certificate D2 supplies an explicit table of 795 witnesses.)

Assembly. Let u^0 be the base offset vector with $u^0_e = 4$ on the first $3s$ positions (forming s blocks), $u^0_e = 5$ on r_0 of the remaining positions, and $u^0_e = 4$ elsewhere, and set $H_0 = F(u^0)$. By stage coverage, there exist s admissible grid values $x_0, \ldots, x_{s-1} \in X$ such that

$$\sum_{i=0}^{s-1} (9/g)\, 10^{3i}\, x_i \equiv \frac{t - H_0}{g} \pmod{M}.$$

Lifting each $x_i = d_{i,1} + 11 d_{i,2}$ to a balanced triple on block i produces an offset vector $u \in \{0, \ldots, 8\}^L$ with $\sum_e u_e = 4L + r_0 = K$ and $H := F(u) = H_0 + g \sum_{i=0}^{s-1} (9/g)\, 10^{3i}\, x_i \equiv t \pmod{m}$. Then $N = B + 10^d \cdot H$ has all digits nonzero, $\mathrm{len}(N) = d + L \geq 37 \geq 2$, and $\sigma(N) = \sigma(B) + L + K = n$. Moreover, $N \equiv 0 \pmod{R}$ (since $R \mid B$ and $R \mid 10^d$) and $N \equiv B + 10^d t \equiv 0 \pmod{m}$; since $\gcd(R, m) = 1$, we conclude $n = Rm \mid N$. □

6 Proof of Theorem 1

Let $m \geq 2$ be coprime to 10, $g = \gcd(9, m)$, $M = m/g$, and $s = \lceil (M-1)/2 \rceil$. Fix a positive integer $L \geq 3s$, an integer $\sigma \in [L + 3s,\, 9L - 3s]$, and a residue $r \equiv \sigma \pmod{g}$. Set $K = \sigma - L$, so $K \in [3s,\, 8L - 3s]$.

If $M = 1$ (that is, $m \in \{3, 9\}$ and $g = m$), then $s = 0$ and $K \in [0,\, 8L]$. Any offset vector $u \in \{0, \ldots, 8\}^L$ with $\sum_e u_e = K$ (obtained greedily by filling offsets up to 8) gives an L-digit nonzero-digit integer $N = F(u)$ with $\sigma(N) = L + K = \sigma$ and $N \equiv \sigma \equiv r \pmod{m}$.

Assume now that $M \geq 3$, so $s = \lceil (M-1)/2 \rceil \geq 1$ and $L \geq 3s \geq 3$.

(i) Offset distribution. Initialize $u^0_e = 1$ on the $3s$ active positions $\{0, \ldots, 3s-1\}$ (which uses $3s$ units of sum) and $u^0_e = 0$ on the remaining $L - 3s$ positions $\{3s, \ldots, L-1\}$. We must distribute the remaining $K - 3s \geq 0$ units across the L positions while keeping $u^0_e \in \{1, \ldots, 7\}$ for $e \in \{0, \ldots, 3s-1\}$ and $u^0_e \in \{0, \ldots, 8\}$ for $e \in \{3s, \ldots, L-1\}$. Each of the $3s$ active positions can absorb up to 6 additional units (from 1 up to 7), and each of the $L - 3s$ free positions can absorb up to 8 additional units (from 0 up to 8). The total remaining capacity is

$$6(3s) + 8(L - 3s) = 8L - 6s.$$

Because $K \leq 8L - 3s$, we have $K - 3s \leq 8L - 6s$, so a greedy allocation produces a base vector u^0 with $\sum_{e=0}^{L-1} u^0_e = K$ and $u^0_e \in \{1, \ldots, 7\}$ for all $e \in \{0, \ldots, 3s-1\}$.

(ii) Residue coverage. Let $H_0 = F(u^0)$. Since $H_0 \equiv L + K = \sigma \equiv r \pmod{g}$, the integer $(r - H_0)/g$ is well-defined modulo M. Because $u^0_e \in \{1, \ldots, 7\}$ on every position of the s disjoint blocks $\{3i, 3i+1, 3i+2\}$ $(0 \leq i < s)$, the minimal moves $U_{\min} = \{\pm 1, \pm 10\}$ are admissible on each block. As shown in §4, for $M \geq 3$ each stage set $B^*_i(U_{\min})$ has cardinality at least 3 (so $w \geq 2$) and every nonzero element is a unit modulo M. By Theorem 5, the s-fold sumset $B^*_0(U_{\min}) + \cdots + B^*_{s-1}(U_{\min})$ has size at least $\min(M, 2s+1) = M$, covering $\mathbb{Z}/M\mathbb{Z}$. Selecting moves $x_i \in U_{\min} \cup \{0\}$ for $0 \leq i < s$ with $\sum_{i=0}^{s-1} (9/g)\, 10^{3i}\, x_i \equiv (r - H_0)/g \pmod{M}$ and applying the corresponding balanced triples in $\{-1, 0, 1\}^3$ yields an offset vector $u \in \{0, \ldots, 8\}^L$ of sum K such that $N = F(u)$ has L nonzero digits, $\sigma(N) = \sigma$, and $N \equiv r \pmod{m}$. □

Remark 6. One cannot uniformly replace $s = \lceil (M-1)/2 \rceil$ in the minimal-move argument by $\lceil (M-1)/4 \rceil$: when $M \in \{9, 11\}$, the congruence $10 \equiv \pm 1 \pmod{M}$ collapses $U_{\min} \bmod M$ to $\{\pm 1\}$, leaving $|B^*_i(U_{\min})| = 3$ $(w = 2)$ so that 4 stages (for $M = 9$) and 5 stages (for $M = 11$) are

genuinely required by minimal moves. An auxiliary machine check over all $m \leq 200$ coprime to 10 at minimal non-vacuous lengths confirms Theorem 1 on every residue and digit sum in the stated band; this computation is not part of the certified development.

7 Method window and edge behavior

We clarify the role of the endpoints of the interval $[L+3s,\, 9L-3s]$ in Theorem 1 (where $L \geq 1$ is assumed). To operate s independent minimal-move exchange blocks simultaneously without risking frozen poles, every one of the $3s$ active positions must lie in $\{1, \ldots, 7\}$. Having at least $3s$ positions with $u_e \geq 1$ forces $K = \sum_e u_e \geq 3s$, i.e. $\sigma \geq L+3s$; symmetrically, having at least $3s$ positions with $u_e \leq 7$ forces $8L - K = \sum_e (8 - u_e) \geq 3s$, i.e. $\sigma \leq 9L - 3s$. In this precise sense, $[L+3s,\, 9L-3s]$ is the exact feasibility window of our s-stage non-frozen block construction.

We do *not* claim that residues outside $[L+3s,\, 9L-3s]$ are mathematically unrealizable: individual moduli m and sums σ near the edges can admit witnesses via denser moves from X or asymmetric position choices. Only at the extreme endpoints $\sigma = L$ (the repunit $11\cdots1$) and $\sigma = 9L$ (the number $99\cdots9$) does the fibre collapse to a single integer with no residue freedom at all. Inside $[L+3s,\, 9L-3s]$, disjointness of the s position triples prevents any interference between stages regardless of the multiplicative order of 10 modulo M.

8 Formal verification

It is important to distinguish what is kernel-verified in Lean 4 from what belongs to the mathematical presentation of this paper:

- *Kernel-verified in Lean 4 (v4.35.0, mathlib v4.35.0) [7]:* the complete characterization of Murthy's conjecture (Theorem 2), all five necessity lemmas, the fixed-sum smooth-exchange machinery feeding the existence proof for $n > 200$, all six finite certificates (A, B, C, D1, D2, and E), and the OEIS sequence-value specification (`IsLeast` for non-exceptional indices, vanishing on exceptional indices, and exact values on displayed OEIS anchors).

- *Proved in this paper only:* the general two-parameter band theorem for arbitrary $\sigma \in [L+3s,\, 9L-3s]$ using minimal moves (Theorem 1, §6) and the quantitative summary (Corollary 3).

In the main file `A077754.lean`, the headline theorem is formalized over $n \in \mathbb{N}$ as the equivalence between the existence of $N \in \mathbb{N}$ satisfying $n \mid N$, allNonzero(N), $\sigma_{10}(N) = n$, and (Nat.digits 10 N).length ≥ 2, and the condition $n \notin \{1, 2, 5, 11\} \cup \{m \in \mathbb{N} : 10 \mid m\}$. Table 1 records the correspondence between results in the text and declarations in the Lean artifact.

In the companion module `A077754Val.lean`, the OEIS sequence itself is defined via `a077754_val`: `a077754_val_spec` certifies that for every non-exceptional n, `a077754_val` n is the least element (`IsLeast`) of the set of admissible multiples, while vanishing exactly on the exceptional set. The displayed OEIS terms are kernel-certified value anchors: $a(3) = 12$, $a(4) = 112$, $a(6) = 24$, $a(7) = 133$, $a(8) = 152$, $a(9) = 18$, and $a(12) = 48$ (each verified together with minimality by scanning all positive multiples $n \cdot j < a(n)$), alongside $a(n) = 0$ at $n \in \{1, 2, 5, 10, 11, 20, 30, 40\}$.

Running `lake build` compiles 9052 jobs with 0 errors and no `sorry` across four Lean modules (approximately 2600 lines in `A077754.lean`, 267 lines in `A077754Cert.lean`, 805 lines in `CornerTable.lean`, and 319 lines in `A077754Val.lean`). Every finite certificate is evaluated inside the Lean kernel via `decide`, avoiding `native_decide` so that the axiom report for `A077754` lists

Paper item	Lean declaration	Role in formalization
Theorem 2	`A077754`	Full iff characterization of Murthy's conjecture
§2 ($n = 1, 2, 5, 10k, 11$)	`exc_1`, `exc_2`, `exc_5`, `exc_10`, `exc_11`	Necessity of the exceptional set
§5 ($n \leq 200$, Cert. A)	`existence_small`	Exhaustive witness table for $n \leq 200$
§5 (inductive / low block)	`existence_step`	Reduction $n = Rm$ via low-order block
§5 (coprime factor m)	`existence_Rm`	High-fibre existence modulo m ($\gcd(m, 10) = 1$)
§3 (exchange grid X)	`Xgrid_card`	Grid structure and cardinality check
§4 (Cert. C, smooth units)	`Xsmooth25_sub`, `Xsmooth235_sub`	Inclusion and counts of smooth grid units
§5 (Cert. B, $M \leq 184$)	`stage_cover_small_aux'`	Bit-parallel stage coverage for $m \leq 1656$
§5 (Cert. D2, corners)	`corner_witness`	795 explicit witnesses for $m \in \{11, 19, 23\}$, $L \in \{7, 8, 9\}$
§8 (OEIS sequence $a(n)$)	`a077754_val`, `a077754_val_spec`	Definition of $a(n)$ and `IsLeast` specification
§8 (Cert. E, zero anchors)	`a077754_val_zero_*`	Certified zeros at displayed exceptional indices

Table 1: Correspondence between mathematical statements and Lean 4 declarations.

only `propext`, `Classical.choice`, and `Quot.sound`. For the sumset coverage certificate B, the kernel evaluates a bit-parallel residue bitmap in the dedicated module `A077754Cert.lean`, and a bridge lemma in `A077754.lean` transports the bitmap verdict to `Finset` coverage.

Artifact. This paper, the Lean formalization (modules `A077754`, `A077754Cert`, `CornerTable`, `A077754Val`), the independent verifier script, and the pinned build files are archived together at `https://doi.org/10.5281/zenodo.23879087`.

9 Remarks

- *Alternative asymptotic tools.* Let $Z_j = (10^j - 1)/9$ denote the repunit of j ones. A two-block expression of the form $Z_a \cdot 10^k + (10^k - Z_r)$ (with $1 \leq r \leq k$, which consists of a ones, followed by $k - r$ nines, $r - 1$ eights, and a final nine, hence all digits nonzero and controllable digit sum) can be combined with Konyagin's results on arithmetic properties of integers with missing digits [4] to treat sufficiently large n. The present proof replaces heavy exponential-sum inputs by Chowla's elementary theorem together with explicit finite certificates that close the threshold down to $n = 1$.

- *Other bases.* In a general base $b \geq 4$, the same fixed-length fibre and balanced triple exchange mechanism applies directly (with the prime factors of b playing the role of $\{2, 5\}$, and digits in $\{1, \ldots, b - 1\}$ leaving $b - 3 \geq 1$ non-frozen interior values $\{2, \ldots, b - 2\}$ (at least two for $b \geq 5$)). Base $b = 2$ degenerates completely since 1 is the unique nonzero digit, and base $b = 3$ has no non-frozen interior digit either (its nonzero digits $\{1, 2\}$ are both frozen poles), so base 3 requires a redesigned multi-position exchange gadget rather than single position-triple offsets.

- *Role of the dense exchange grid.* Balanced digit exchanges and Chowla's 1935 sumset inequality are classical tools, but using minimal moves alone ($w = 2$) would require $L \approx 3M/2$ digits, exceeding the available linear digit budget $L \approx n/5$ when $g = 1$ and $n = M$. Enlarging the single-block grid X to 60 nonzero values yields 28 smooth units when $g = 1$ and 16 when $g \in \{3, 9\}$, reducing the stage slope to $3/28 < 1/5$ (respectively $3/16 < 3/5$) and pushing the analytic threshold down to $M \leq 184$, well within reach of kernel-checked bitmap evaluation.

A Reproducible machine certificates

Six finite certificates are used in the formalization; all six are kernel-checked by `decide` inside the Lean artifact (§8), and certificates A, B, and C are independently re-derived in seconds by the standalone script `A077754-ANNEXE.py` (Python standard library, around 90 lines):

- *Certificate A (small n witnesses):* an explicit admissible multiple N for each of the 176 non-exceptional positive integers $n \leq 200$, found by breadth-first search over states (residue mod n, digit sum, length).
- *Certificate B (small-M stage coverage):* in `A077754Cert.lean`, bit-parallel sumset evaluation verifies that for every $m \leq 1656$ coprime to 10 with $2 \leq M \leq 184$ and $m \notin \{11, 19, 23\}$, the smooth move set $U_{2,3,5}$ (28 values, used when $g = 1$) or $U_{2,5}$ (16 values, used when $g \in \{3, 9\}$) covers all residues in the appropriate mod g coset of $\mathbb{Z}/m\mathbb{Z}$ within $k \leq \lfloor m/27 \rfloor + 1$ stages. (The Python script also runs a uniform check showing that every $M \leq 600$ coprime to 10 is covered in at most 6 stages, with worst case $M = 333$ where $10^3 \equiv 1 \pmod{333}$.)
- *Certificate C (smooth grid units):* verification on the 61-element grid X that X contains the 16 $\{2, 5\}$-smooth nonzero values $\pm\{1, 2, 4, 8, 10, 20, 32, 40\}$ (used for $g \in \{3, 9\}$) together with the 12 additional $\{2, 3, 5\}$-smooth values $\pm\{3, 9, 12, 18, 24, 30\}$, giving 28 smooth units for $g = 1$.
- *Certificate D1 (two-stage coverage for $\{11, 19, 23\}$):* a finite stage-coverage check (not a witness table) confirming that $k = 2$ stages of smooth moves from $U_{2,3,5}$ (which applies since $g = \gcd(9, m) = 1$ for all three moduli) cover $\mathbb{Z}/m\mathbb{Z}$ for $m \in \{11, 19, 23\}$ (in fact $k = 1$ already covers $m = 19$ and $m = 23$).
- *Certificate D2 (corner witness table):* in `CornerTable.lean` and `A077754.lean`, an explicit table of 795 digit witnesses covering all residues modulo m for $m \in \{11, 19, 23\}$, $L \in \{7, 8, 9\}$, and $r_0 \in \{0, 1, 2, 3, 4\}$. As noted in §5, $n > 200$ forces $L \geq 37$ in the main theorem, so D2 is used solely to close the standalone helper lemma at its stated lower bound $L \geq \lfloor m/5 \rfloor$.
- *Certificate E (OEIS value anchors):* in `A077754Val.lean`, kernel verification of the exact sequence values $a(3) = 12$, $a(4) = 112$, $a(6) = 24$, $a(7) = 133$, $a(8) = 152$, $a(9) = 18$, and $a(12) = 48$ (each paired with a kernel scan over $n \cdot j < a(n)$ proving minimality, distinguishing e.g. the minimal value $a(7) = 133$ from non-minimal existential witnesses), together with $a(n) = 0$ for $n \in \{1, 2, 5, 10, 11, 20, 30, 40\}$.

Use of AI tools

Generative AI tools assisted in blind OEIS transcription, adversarial auditing, proof sketching, and text review, within a verification methodology designed and supervised by the author. The characterization of OEIS A077754 (Theorem 2) and its sequence-value specification are verified from standard axioms only in the Lean 4 kernel, with the Lean source and the independent Python certificate script archived in the companion record. The author selected the problems and proof strategies, arbitrated all discrepancies, verified OEIS fidelity, and takes full responsibility for this work.

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Competing interests

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